Adam's Setup Guide

AI-assisted game development on your RTX 5080

What You Have

An RTX 5080 with 16GB VRAM and CUDA acceleration. This runs AI models locally — no cloud, no subscription, no rate limit. You can run a 27-billion-parameter model that writes working code at reading speed.

Step 1 — Roblox Studio

Download from **roblox.com/create**. Click Start Creating. Free account required.

Verify: Open Roblox Studio. You should see the template picker (Baseplate, Platformer, Racing, etc.). Pick Baseplate.

Step 2 — Ollama

This runs AI models on your GPU. No internet needed after download.

Install: Open PowerShell as Administrator and run:

```
winget install Ollama.Ollama
```

Pull your model:

```
ollama pull hf.co/OrcaRouter/OrcaSAQ-2-27B-gguf
```

This downloads 12GB. It fits in your 16GB VRAM. It writes working code (70.0 SWE-bench).

Verify:

```
ollama run hf.co/OrcaRouter/OrcaSAQ-2-27B-gguf "Write a Lua function that prints hello"
```

If it responds with Lua code, your GPU is doing AI. Type /bye to exit.

Check GPU is being used:

```
ollama ps
```

Look at the PROCESSOR column. It should say 100% GPU. If it says 100% CPU, your NVIDIA driver needs updating — go to nvidia.com/drivers, download the latest, install, restart.

Gotcha: Ollama's default context window is 4096 tokens. For game development you need more. Create a Modelfile:

```
FROM hf.co/OrcaRouter/OrcaSAQ-2-27B-gguf
PARAMETER num_ctx 32768

ollama create orcasaq-32k -f Modelfile
```

Now use orcasaq-32k as your model name. You have 32K context — enough to hold your entire game script.

Step 3 — Rojo

Rojo syncs code files from your computer into Roblox Studio automatically. Write code in VS Code, save, and it appears in Studio.

Install:

```
mkdir C:\tools -Force; cd C:\tools curl -L -o rojo.zip "https://github.com/rojo-rbx/rojo/releases/download/v7.7.0/rojo-7.7.0-windows-x86_64.zip" Expand-Archive rojo.zip -DestinationPath rojo
```

Verify:

```
C:\tools\rojo\rojo.exe --version
```

Should print: Rojo 7.7.0

Gotcha: You also need the Rojo plugin inside Roblox Studio. Open Studio, go to Plugins → Manage Plugins → search for Rojo → Install.

Step 4 — Connect Hermes to Your Local Model

Hermes is already installed. Point it at Ollama:

In Hermes config or chat settings:
Model endpoint: http://localhost:11434/v1
Model name: orcasaq-32k
Max tokens: 4096
Temperature: 0.2

Verify: Ask Hermes to write a Lua function. If it responds, you're running AI on your own GPU with zero API cost.

Gotcha: If Hermes can't connect, make sure Ollama is running (ollama serve in a terminal). If the model is slow, check that the PROCESSOR column in ollama ps says GPU, not CPU.

Step 5 — VS Code

```
winget install Microsoft.VisualStudioCode code --install-extension sumneko.lua code --install-extension rojo-rbx.rojo
```

Verify: Open VS Code, create a new file, type print("hello"), save it as test.lua. You should see Lua syntax highlighting.

Step 6 — Git

```
winget install Git.Git
```

Verify: git --version

Step 7 — Get the Game Code

The starter Luau code for your game is already written. Your dad will share the repo or the files. The five modules:

File	What It Does
GameConfig.lua	Game constants and settings
ShadowLedger.lua	Tracks what players do
TheAnswer.lua	The boss fight logic
HallOfSupper.lua	The starting room
ServerMain.lua	Connects everything

Put them in a folder with a default.project.json (Rojo's config). Run rojo serve, connect from Studio, and the code syncs.

Free API Keys (Optional)

Sometimes you want a bigger model than your GPU can run. These are free:

Service	Sign Up At
OpenRouter	openrouter.ai
Groq	console.groq.com
NVIDIA Build	build.nvidia.com
Cerebras	cloud.cerebras.ai

Put keys in a .env file. One key per line: KEY_NAME=value

Your Daily Workflow

Every day, same steps:

1. Open a terminal in your game folder
2. Run: rojo serve default.project.json
3. Open Roblox Studio → Rojo plugin → Connect
4. Ask Hermes to write code: "Write a stamina system for a souls-like game"
5. Hermes writes the Lua file — it syncs to Studio instantly
6. Press Play in Studio to test
7. If it breaks, paste the error back into Hermes: "Fix this: [error]"
8. Repeat until it works
9. Publish: File → Publish to Roblox

Common Mistakes

Mistake	Fix
Ollama says 100% CPU	Update your NVIDIA driver. Restart Ollama completely.
Rojo won't connect	Make sure rojo serve is running in a terminal. Check the port number matches.
Script doesn't run in Studio	Check the Output panel (View → Output). The error tells you what's wrong.
Hermes gives empty responses	Your model consumed the token budget on thinking. Add thinking: disabled to your
Everything is slow	Check ollama ps. If it says CPU, fix your GPU driver. If it says GPU but slow, reduce

How You Get Paid

Roblox pays 70% of what players spend. $0.0035 per Robux. Minimum payout: 30,000 Robux (~$105).

Free-to-play game → Game Pass (99 Robux: double score) → Dev Product (49 Robux: extra life) → Premium Payouts (passive income from engagement time). 430 pass sales = first payout.

Post your launch clip. Your development loop is instant, free, and unlimited.

All Links

Resource	URL
Roblox Studio	roblox.com/create
Roblox Docs	create.roblox.com/docs
Luau Language	luau.org
Rojo	rojo.space
Ollama	ollama.com
OrcaSAQ-2 Model	huggingface.co/OrcaRouter/OrcaSAQ-2-27B-gguf
GameDev OS (free assets)	github.com/Scenario-Games/GameDev-OS
OpenRouter	openrouter.ai
Groq	console.groq.com
NVIDIA Build	build.nvidia.com
Free Game Assets	itch.io/game-assets/free
Free Sound Effects	freesound.org
Free Music	freepd.com
Hermes Docs	hermes-agent.nousresearch.com/docs

Build the map. The territory follows.